ASSIGNMENT No: 04 (BI)
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Title: Molecular Docking and Virtual Screening
Code:

```
import pandas as pd
import matplotlib.pyplot as plt

print("MOLECULAR DOCKING AND VIRTUAL SCREENING")
print("-"*60)

data={

"Ligand":["Ligand_A","Ligand_B","Ligand_C","Ligand_D","Ligand_E","Ligand_F","Ligand_G","
Ligand_H"],
   "Binding_Affinity":[-5.2,-7.8,-6.1,-9.0,-4.8,-8.2,-6.7,-7.1]
}

df=pd.DataFrame(data)

print("\nDocking Results:")
print(df)

ranked_df=df.sort_values(by="Binding_Affinity")
ranked_df=ranked_df.reset_index(drop=True)
ranked_df["Rank"]=ranked_df.index+1

print("\nRanked Ligands:")
print(ranked_df[["Rank","Ligand","Binding_Affinity"]])

top_candidates=ranked_df.head(3)

print("\nTop 3 Potential Drug Candidates:")
print(top_candidates[["Rank","Ligand","Binding_Affinity"]])

best_ligand=ranked_df.iloc[0]

print("\nBest Candidate:")
print("Ligand:",best_ligand["Ligand"])
print("Binding Affinity:",best_ligand["Binding_Affinity"],"kcal/mol")

plt.figure(figsize=(10,6))
plt.bar(ranked_df["Ligand"],ranked_df["Binding_Affinity"])
plt.xlabel("Ligands")
plt.ylabel("Binding Affinity (kcal/mol)")
plt.title("Molecular Docking Scores")
```

```
plt.xticks(rotation=45)
plt.axhline(y=0,linewidth=1)
plt.tight_layout()
plt.show()

print("\nRESULT")
print("-"*60)
print("Virtual screening completed successfully.")
print("The ligand with the most negative docking score")
print("was identified as the best predicted candidate.")
print("Aditya Rade   BEAD23167")
```

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OUTPUT:

```
Problems 10   Debug Console   Output   Terminal   Ports   Jupyter

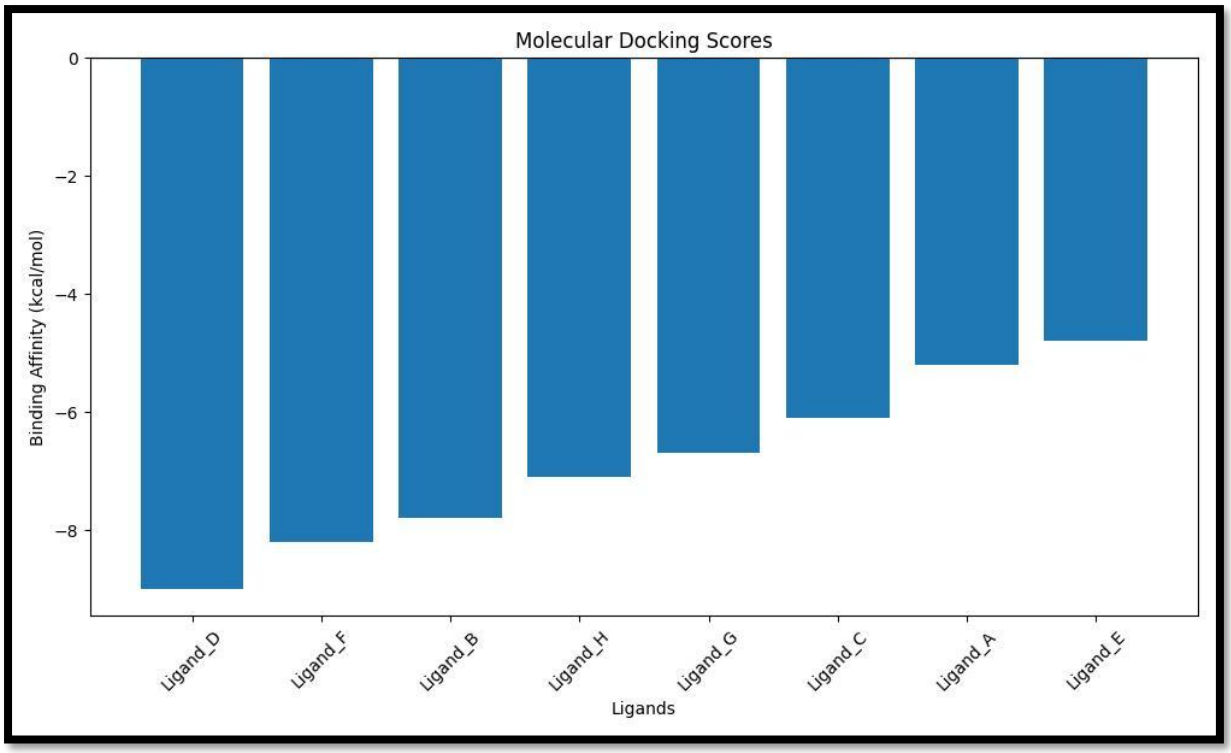
PS C:\Users\adity\OneDrive\Documents\Desktop\Sem 7 study materials\CL - 2> & C:\Users\adity\AppData\Local\Programs\
OneDrive/Documents/Desktop/Sem 7 study materials/CL - 2/Practical-4-BI.py"
MOLECULAR DOCKING AND VIRTUAL SCREENING
------------------------------------------------------------

Docking Results:
     Ligand  Binding_Affinity
0  Ligand_A              -5.2
1  Ligand_B              -7.8
2  Ligand_C              -6.1
3  Ligand_D              -9.0
4  Ligand_E              -4.8
5  Ligand_F              -8.2
6  Ligand_G              -6.7
7  Ligand_H              -7.1

Ranked Ligands:
   Rank    Ligand  Binding_Affinity
0     1  Ligand_D              -9.0
1     2  Ligand_F              -8.2
2     3  Ligand_B              -7.8
3     4  Ligand_H              -7.1
4     5  Ligand_G              -6.7
5     6  Ligand_C              -6.1
6     7  Ligand_A              -5.2
7     8  Ligand_E              -4.8
```

```
Top 3 Potential Drug Candidates:
   Rank    Ligand  Binding_Affinity
0     1  Ligand_D             -9.0
1     2  Ligand_F             -8.2
2     3  Ligand_B             -7.8

Best Candidate:
Ligand: Ligand_D
Binding Affinity: -9.0 kcal/mol
```


Molecular Docking Scores
Binding Affinity (kcal/mol)
0
−2
−4
−6
−8
Ligand_D
Ligand_F
Ligand_B
Ligand_H
Ligand_G
Ligand_C
Ligand_A
Ligand_E
Ligands